PADHAKU
Ganita Manjari Part II · Class 9 Mathematics



CHAPTER 13

Two Variables, One Line

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡
answers are samples: write your own.

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Exercise Set 13.1

1 **Write a linear equation in two variables in which a = 3, b = 0 and c = −1/5.**

✔ ANSWER

Standard form: ax + by + c = 0

Put a = 3, b = 0, c = −1/5:

3x + 0×y − 1/5 = 0

(Multiplying by 5: 15x + 0×y − 1 = 0.)

2 **Complete the table after writing each equation in standard form ax + by + c = 0:**
y − 15 = 2x; 3y − 2x = 0; 5x = 3y; x = 8; 3y = 1

✔ ANSWER

Equation	Standard form	Coeff. of x	Coeff. of y	Constant
y − 15 = 2x	2x − y + 15 = 0	2	−1	15
3y − 2x = 0	−2x + 3y + 0 = 0	−2	3	0
5x = 3y	5x − 3y + 0 = 0	5	−3	0
x = 8	x + 0×y − 8 = 0	1	0	−8
3y = 1	0×x + 3y − 1 = 0	0	3	−1

(You may also multiply a whole equation by −1, e.g. −2x + y − 15 = 0; then all three numbers change sign.)

3 (i) **The cost of a notebook (₹t) is twice the cost of a pen (₹p). Charlie wrote t = 2p, Meera wrote p = 2t. Which is correct?**

✔ ANSWER

Notebook cost = 2 × pen cost, so t = 2 × p.

Charlie's t = 2p is correct.

Meera's p = 2t would mean the pen costs twice the notebook.

3 (ii) **Two Indian batsmen together scored 176 runs. Manisha wrote x + y = 176 (x, y = runs of each batsman). Is this correct?**

✔ ANSWER

Runs of first + runs of second = total runs.

Yes, x + y = 176 is correct (x and y are whole numbers).

🤔 Think and Reflect · 13.2.1 Algebraic Meaning of Solution

? **Are (1, 9/2), (0, 6), (4, 1) and (1/2, 21/4) solutions of 3x + 2y = 12? Can you find any other solution? How many solutions can you find?**

✔ ANSWER

(1, 9/2): 3×1 + 2×9/2 = 3 + 9 = 12 → **yes**

(0, 6): 0 + 12 = 12 → **yes**

(4, 1): 12 + 2 = 14 ≠ 12 → **no**

(1/2, 21/4): 3/2 + 21/2 = 24/2 = 12 → **yes**

Other solutions: (2, 3), (4, 0), (−2, 9), (6, −3)...

For every value of x we get a value of y, so there are **infinitely many solutions**.

🤔 Think and Reflect · 13.2.2 Geometric Visualisation of Solutions

? **Ranju, Meena, Farhan and Toshi conclude: whenever c = 0, the line ax + by + c = 0 always passes through the origin, whatever a and b are. Do you agree?**

✔ ANSWER

Yes. If c = 0 the equation is ax + by = 0.

Put x = 0, y = 0: a×0 + b×0 = 0, which is true for any a and b.

So (0, 0) is always a solution, i.e. the line passes through the origin.

Exercise Set 13.2

1 **Verify if the ordered pair (4, 3) is a solution of $5x - 6y = 2$. Explain your reasoning.**

✔ ANSWER

Put $x = 4$, $y = 3$: $5\times4 - 6\times3 = 20 - 18 = 2 =$ RHS

Yes, (4, 3) is a solution, because it makes LHS = RHS.

2 (i) **Find any two solutions of $7x - 3y = 21$.**

✔ ANSWER

$x = 0$: $-3y = 21 \rightarrow y = -7$

$y = 0$: $7x = 21 \rightarrow x = 3$

(0, −7) and (3, 0) (others: (6, 7), ...)

2 (ii) **Find any two solutions of $2x + 3y = 5$.**

✔ ANSWER

$x = 1$: $2 + 3y = 5 \rightarrow y = 1$

$x = 4$: $8 + 3y = 5 \rightarrow y = -1$

(1, 1) and (4, −1) (others: (0, 5/3), (5/2, 0), ...)

3 **$2mx + 3y = 7$ and $4x + ny = -10$. If (2, −1) is the solution of both equations, find m and n.**

✔ ANSWER

First equation, $x = 2$, $y = -1$:

$2m(2) + 3(-1) = 7$

$4m - 3 = 7 \rightarrow 4m = 10 \rightarrow m = 5/2$

Second equation:

$4(2) + n(-1) = -10$

$8 - n = -10 \rightarrow n = 18$

m = 5/2, n = 18

4 (i) **Find two solutions in different quadrants of 5x + 3y = 7 and name the quadrants.**

✔ ANSWER

$x = -1$: $-5 + 3y = 7 \rightarrow y = 4 \rightarrow (-1, 4)$, **Quadrant II**

$x = 2$: $10 + 3y = 7 \rightarrow y = -1 \rightarrow (2, -1)$, **Quadrant IV**

(Also (1, 2/3) in Quadrant I.)

4 (ii) **5x − 3y = 7**

✔ ANSWER

$x = 2$: $10 - 3y = 7 \rightarrow y = 1 \rightarrow (2, 1)$, **Quadrant I**

$x = -1$: $-5 - 3y = 7 \rightarrow y = -4 \rightarrow (-1, -4)$, **Quadrant III**

4 (iii) **−5x + 3y = 7**

✔ ANSWER

$x = 1$: $-5 + 3y = 7 \rightarrow y = 4 \rightarrow (1, 4)$, **Quadrant I**

x = −2: 10 + 3y = 7 → y = −1 → (−2, −1), **Quadrant III**

4 (iv) **−5x − 3y = 7 (then verify by drawing the lines on graph paper).**

✔ ANSWER

x = 1: −5 − 3y = 7 → y = −4 → (1, −4), **Quadrant IV**

x = −2: 10 − 3y = 7 → y = 1 → (−2, 1), **Quadrant II**

Verification: plot the two points of each equation and join them; each line also passes through the x- and y-intercepts, e.g. (7/5, 0) and (0, 7/3) for 5x + 3y = 7.

5 **Graph of 3x − 7y = 21 passes through A (7, 0) and B (0, −3). Does C (2, 3) lie on the line? Does it satisfy the equation? Can points that do not lie on the line satisfy the equation?**

✔ ANSWER

From the graph, C (2, 3) is **not on the line**.

Check: 3×2 − 7×3 = 6 − 21 = −15 ≠ 21, so it **does not satisfy** the equation.

No. Every solution is a point on the line, and every point on the line is a solution. A point off the line can never satisfy the equation.

6 (i) **True or False: A linear equation in two variables has only one solution.**

✔ ANSWER

False. It has infinitely many solutions (one for every value of x).

6 (ii) **The graph of a linear equation in two variables always passes through the origin.**

✔ ANSWER

False. Only when c = 0. E.g. x + y = 5: (0, 0) gives 0 ≠ 5.

6 (iii) **A linear equation in two variables can never have rational solutions.**

✔ ANSWER

False. E.g. 3x + 2y = 12 has the solution (2, 3) and (1, 9/2).

6 (iv) **x = 3 is a valid linear equation in two variables.**

✔ ANSWER

True. It can be written x + 0×y − 3 = 0 (a = 1, b = 0, c = −3).

6 (v) **The equation 2x + 3y = 7 has infinitely many solutions.**

✔ ANSWER

True. Every linear equation in two variables has infinitely many solutions, e.g. (2, 1), (5, −1), (−1, 3)...

6 (vi) **The point (1, 2) is a solution of 2x + 3y = 7.**

✔ ANSWER

$2\times1 + 3\times2 = 2 + 6 = 8 \neq 7$

False.

7 (i) **Compare the solutions of 3x + 4y = 7 and 6x + 8y = 14. Argue that they have the same set of solutions.**

✔ ANSWER

$6x + 8y = 14$ is just $2 \times (3x + 4y = 7)$.

If (p, q) satisfies $3p + 4q = 7$, then $6p + 8q = 2(3p + 4q) = 2\times7 = 14$. So it is a solution of the second equation.

If $6p + 8q = 14$, divide by 2: $3p + 4q = 7$. So it is a solution of the first.

Example: (1, 1): $3 + 4 = 7$ and $6 + 8 = 14$.

Hence both equations have the same solutions (the same line).

7 (ii) **Show that ax + by = c and kax + kby = kc (k ≠ 0) have the same set of solutions.**

✔ ANSWER

If ap + bq = c, multiply by k: kap + kbq = kc. So (p, q) solves the second equation.

If kap + kbq = kc, divide by k (allowed since k ≠ 0): ap + bq = c. So (p, q) solves the first.

So every solution of one is a solution of the other: same set of solutions.

🤔 Think and Reflect · 13.3.1 Understanding Slope

1 **Line AB passes through A (2, 0) and B (4, 3) (slope 3/2). Find the y-coordinate of the point on AB whose x-coordinate is 8.**

✔ ANSWER

From A, run = 8 − 2 = 6

Rise = slope × run = 3/2 × 6 = 9

y = 0 + 9 = 9

(8, 9)

2 **What is the x-coordinate of the point on AB whose y-coordinate is 6?**

✔ ANSWER

From A, rise = 6 − 0 = 6

Run = rise ÷ slope = 6 ÷ 3/2 = 4

x = 2 + 4 = 6

(6, 6)

🤔 Think and Reflect · 13.3.2 Slope-Intercept Form

? **Use the points A (0, 3), B (1, −2) and C (−1, 8) to verify that the slope of y = −5x + 3 is −5.**

✔ ANSWER

A and B: (−2 − 3)/(1 − 0) = −5/1 = −5

A and C: (8 − 3)/(−1 − 0) = 5/(−1) = −5

B and C: (8 − (−2))/(−1 − 1) = 10/(−2) = −5

Each pair gives slope −5.

📝 Exercise Set 13.3

1 **Ski hills (height, base): A: 60 m, 70 m; B: 60 m, 110 m; C: 80 m, 100 m.**

Rank the hills in order of steepness, from least to greatest.

✔ ANSWER

Slope = rise ÷ run

A: 60/70 = 6/7 ≈ 0.86

B: 60/110 = 6/11 ≈ 0.55

C: 80/100 = 0.8

Least to greatest: B, C, A

2 **A ramp rises 2.5 m over a run of 4 m. Find the slope.**

✔ ANSWER

Slope = rise/run = 2.5/4 = 25/40

= 5/8 = 0.625

3 (i) **Find the slope of line l: (i) l is a horizontal line (parallel to the x-axis).**

✔ ANSWER

y does not change, rise = 0

Slope = 0

3 (ii) **(ii) l passes through the origin and goes down to the right, through (4, −4) on the grid.**

✔ ANSWER

Slope = (−4 − 0)/(4 − 0)

= −1

3 (iii) **(iii) l passes through the marked points (−4, −2) and (2, 2) on the grid.**

✔ ANSWER

Rise = 2 − (−2) = 4

Run = 2 − (−4) = 6

Slope = 4/6 = **2/3**

3 (iv) **(iv) l is a vertical line (parallel to the y-axis).**

✔ ANSWER

x does not change, run = 0, division by 0

Slope is undefined

4 **A ramp's slope must be 1 cm rise for 12 cm horizontal length. The stairs are 18 cm high. What should be the horizontal length of the ramp?**

✔ ANSWER

rise/run = 1/12

18/run = 1/12

run = 18 × 12 = 216

Horizontal length = 216 cm (2.16 m)

🤔 Think and Reflect · 13.4 Pair of Linear Equations in Two Variables

? **1 paratha and 2 bowls of dahi cost ₹90; 2 parathas and 7 bowls of dahi cost ₹260. Find the cost of one paratha and one bowl of dahi.**

✔ ANSWER

Let paratha = ₹p, dahi = ₹d.

p + 2d = 90 ...(1)

2p + 7d = 260 ...(2)

Double (1): 2p + 4d = 180 ...(3)

(2) − (3): 3d = 80 → d = 80/3

p = 90 − 2 × 80/3 = 90 − 160/3 = 110/3

Paratha = ₹110/3 ≈ ₹36.67, dahi = ₹80/3 ≈ ₹26.67

Reasoning: doubling the first order gives 2 parathas + 4 dahi = ₹180; the second order has 3 more bowls of dahi and costs ₹80 more, so 3 bowls cost ₹80.

? Science exhibition: 2x + 3y = 600 and 3x + 2y = 700 (x = adult ticket, y = child ticket). Why will a solution give the answer we seek?

✔ ANSWER

A solution (x, y) makes both groups' bills correct at the same time, so x and y are the actual ticket prices.

Solving: add: 5x + 5y = 1300 → x + y = 260

Subtract (2nd − 1st): x − y = 100

So x = 180, y = 80

Adult ticket ₹180, child ticket ₹80.

📝 Exercise Set 13.4

1 Movie tickets ₹x, snack box ₹y. (i) Day 1: 2 tickets + 3 snack boxes = ₹850. (ii) Day 2: 4 tickets + 1 snack box = ₹1100.

✔ ANSWER

2x + 3y = 850

4x + y = 1100

2 Taxi: fixed charge ₹x, ₹y per km. (i) Sahil: 6 km, ₹122. (ii) Meena: 8 km, ₹160.

✔ ANSWER

x + 6y = 122

x + 8y = 160

3 Adult ticket ₹150, child ticket ₹100; 200 people attended; ₹25,000 collected. (x adult, y children's tickets)

✔ ANSWER

Number of people: **x + y = 200**

Money: **150x + 100y = 25000** (or 3x + 2y = 500)

🤔 Think and Reflect · 13.5.1 Algebraic Methods

? Shape puzzle (4 × 4 grid):

Row 1: ▲ ▭ ▭ ▲ = 28

Row 2: ⬢ ▭ ⬢ ▭ = 30

Row 3: ● ▲ ● ● = 18

Row 4: ● ▭ ● ● = 20

Column totals: ?, 30, 23, 22

Find the value of each shape.

✔ ANSWER

Row 1: 2T + 2R = 28 → T + R = 14

Last column: T + R + 2C = 22 → 14 + 2C = 22 → C = 4

Row 3: 3C + T = 18 → 12 + T = 18 → T = 6

Row 4: 3C + R = 20 → R = 8

Row 2: 2H + 2R = 30 → H + 8 = 15 → H = 7

Check columns: 2nd: 8 + 8 + 6 + 8 = 30 ✓; 3rd: 8 + 7 + 4 + 4 = 23 ✓

Triangle = 6, Rectangle = 8, Circle = 4, Hexagon = 7

First column: 6 + 7 + 4 + 4 = **21** (the missing total)

? **In Example 7 (7x − 15y = 2, x + 2y = 3), what if we write y = ½(3 − x)? Would we get the same solution?**

✔ ANSWER

Substitute in 7x − 15y = 2: 7x − 15(3 − x)/2 = 2

Multiply by 2: 14x − 45 + 15x = 4

29x = 49 → x = 49/29

y = ½(3 − 49/29) = ½ × 38/29 = 19/29

Yes, the same solution x = 49/29, y = 19/29.

🤔 Think and Reflect · Elimination Method

? **Solve 9x − 4y = 2000, 7x − 3y = 2000 by eliminating x instead of y, and also by substitution. Assess the pros and cons of each method.**

💡 SAMPLE ANSWER

Eliminating x: (1) × 7: 63x − 28y = 14000; (2) × 9: 63x − 27y = 18000

Subtract: y = 4000

Then 9x = 2000 + 16000 = 18000 → x = 2000

Substitution: from (1), x = (2000 + 4y)/9. Put in (2): 7(2000 + 4y)/9 − 3y = 2000

14000 + 28y − 27y = 18000 → y = 4000, x = 2000

Same answer: x = 2000, y = 4000.

Substitution is easy when a variable has coefficient 1; here it gives fractions. Elimination avoids fractions when coefficients are larger, but needs suitable multipliers.

? **Does a pair of linear equations always have a unique solution? Can you think of a situation where the second variable also gets eliminated?**

💡 SAMPLE ANSWER

No. If one equation is a multiple of the other on the left side, both variables vanish together.

E.g. $x - y = 10$ and $10x - 10y = 100$: multiplying the first by 10 and subtracting gives $0 = 0$ (infinitely many solutions).

E.g. $x - y = 10$ and $10x - 10y = 101$: we get $0 = 1$, which is impossible (no solution).

? How many solutions are there to $10x - 5y = 20$ and $4x - 2y = 8$? Give 3 more examples that lead to $0 = 0$, and find a simple rule.

💡 SAMPLE ANSWER

$10x - 5y = 20 \div 5 \rightarrow 2x - y = 4$

$4x - 2y = 8 \div 2 \rightarrow 2x - y = 4$

Same equation → **infinitely many solutions**

Examples: $x + y = 2$ and $2x + 2y = 4$; $3x - y = 5$ and $6x - 2y = 10$; $x + 2y = 7$ and $3x + 6y = 21$

Rule: $a_1/a_2 = b_1/b_2 = c_1/c_2$ (one equation is k times the other).

? What if one of a_2, b_2, c_2 is zero?

💡 SAMPLE ANSWER

Then we cannot divide by it, so use $a_1 = ka_2$, $b_1 = kb_2$, $c_1 = kc_2$ directly.

If $a_2 = 0$, then a_1 must also be 0, and the other two ratios must be equal. E.g. $2y = 6$ and $4y = 12$ ($a = 0$): infinitely many solutions.

If $c_2 = 0$, then $c_1 = 0$ too: e.g. $x + y = 0$ and $3x + 3y = 0$ have infinitely many solutions; $x + y = 0$ and $x + y = 2$ have none.

If $a_1 \neq 0$ while $a_2 = 0$ (or b's similarly), the equations are not multiples, so the solution is unique (if b's allow it).

? **Does a pair of linear equations always have either a unique solution or infinitely many solutions?**

✔ ANSWER

No. It can have no solution. E.g. $x - y = 10$ and $10x - 10y = 101$: the left sides become equal but the right sides (100 and 101) differ.

? **Give 3 more examples of pairs that have no solution. Find a simple rule.**

💡 SAMPLE ANSWER

$x + y = 3$ and $x + y = 5$

$2x - y = 1$ and $4x - 2y = 7$

$x + 3y = 4$ and $3x + 9y = 10$

Rule: $\mathbf{a_1/a_2 = b_1/b_2 \neq c_1/c_2}$ (left sides are multiples, constants are not).

? **Are the converses of the three statements (unique / infinitely many / no solution) true?**

💡 SAMPLE ANSWER

Yes. The three cases cover every pair and cannot happen together.

- If the pair has a unique solution, the ratios cannot be equal (equal ratios give 0 = 0 or 0 = a non-zero number), so $a_1/a_2 \neq b_1/b_2$.
- If there are infinitely many solutions, it is not case 1 or 3, so $a_1/a_2 = b_1/b_2 = c_1/c_2$.
- If there is no solution, it is not case 1 or 2, so $a_1/a_2 = b_1/b_2 \neq c_1/c_2$.

? **Can there be methods other than elimination and substitution? Describe them.**

💡 **SAMPLE ANSWER**

Yes, for example:

- Graphical method: draw both lines; the meeting point is the solution.
- Comparison: write y (or x) in terms of the other from both equations and set them equal.
- Cross-multiplication formula (you will learn it later).
- Adding/subtracting when coefficients are swapped (like 9x + 7y, 7x + 9y).

⭐ 13.6 Graphical Method · Pairs 1–4

? **For each pair, prepare a table (at least two ordered pairs), plot and draw the lines:**

Pair 1: 2x + y = 6 and x − y = 2

Pair 2: x + y = 4 and 2x + 2y = 8

Pair 3: 3x − 2y = 6 and 6x − 4y = 12

Pair 4: $x - 2y = 4$ and $2x - 4y = 6$

✔ ANSWER

Pair 1: $2x + y = 6 \rightarrow (0, 6), (3, 0)$; $x - y = 2 \rightarrow (0, -2), (2, 0)$.

Lines **intersect** at $(8/3, 2/3)$ (add: $3x = 8$): unique solution.

Pair 2: $x + y = 4 \rightarrow (0, 4), (4, 0)$; $2x + 2y = 8 \rightarrow$ same points. **Coincident** lines: infinitely many solutions.

Pair 3: $3x - 2y = 6 \rightarrow (0, -3), (2, 0)$; $6x - 4y = 12 \rightarrow$ same points. **Coincident**: infinitely many solutions.

Pair 4: $x - 2y = 4 \rightarrow (0, -2), (4, 0)$; $2x - 4y = 6 \rightarrow (0, -1.5), (3, 0)$. **Parallel** lines: no solution.

? **How do we find solutions from the graph? How many solutions when the lines (i) intersect at a point, (ii) are parallel, (iii) are coincident?**

✔ ANSWER

A solution is a point lying on both lines.

(i) Intersect: **one (unique) solution**, the meeting point.

(ii) Parallel: **no solution**.

(iii) Coincident: **infinitely many solutions** (every point of the line).

? **Prove that two lines of equal slope are parallel. (Hint: $y = mx + d_1$ and $y = mx + d_2$)**

✔ ANSWER

1. Let the lines be $y = mx + d_1$ and $y = mx + d_2$ with $d_1 \neq d_2$.
2. Suppose they had a common point (p, q). Then $q = mp + d_1$ and $q = mp + d_2$.
3. Subtracting: $0 = d_1 - d_2$, so $d_1 = d_2$. This contradicts $d_1 \neq d_2$.
4. So the lines have no common point: **they are parallel**. (If $d_1 = d_2$, they are the same line.)

📝 Exercise Set 13.5

1 (i) **The sum of two integers is +5 and their difference is −21. Find the numbers.**

✔ ANSWER

$x + y = 5$, $x - y = -21$

Add: $2x = -16 \rightarrow x = -8$

$y = 5 - (-8) = 13$

The numbers are −8 and 13 ($-8 - 13 = -21$)

1 (ii) **The difference between two numbers is 26 and one number is three times the other. Find the numbers.**

✔ ANSWER

$x - y = 26$, $x = 3y$

3y − y = 26 → y = 13

x = 39

39 and 13

1 (iii) **7 bats and 6 balls cost ₹8880; 3 bats and 5 balls cost ₹4000. Find the cost of each bat and ball.**

✔ ANSWER

7x + 6y = 8880 ...(1); 3x + 5y = 4000 ...(2)

(1) × 5: 35x + 30y = 44400

(2) × 6: 18x + 30y = 24000

Subtract: 17x = 20400 → x = 1200

3(1200) + 5y = 4000 → 5y = 400 → y = 80

Bat ₹1200, ball ₹80

1 (iv) **Taxi: 10 km costs ₹155, 15 km costs ₹220. Find the fixed charge and the charge per km. How much for 25 km?**

✔ ANSWER

Fixed charge ₹x, ₹y per km

x + 10y = 155; x + 15y = 220

Subtract: 5y = 65 → y = 13

x = 155 − 130 = 25

Fixed charge ₹25, ₹13 per km

25 km: 25 + 25 × 13 = 25 + 325 = **₹350**

1 (v) **A fraction becomes 9/11 if 2 is added to both numerator and denominator. If 3 is added to both, it becomes 5/6. Find the fraction.**

✔ ANSWER

Fraction x/y

$(x + 2)/(y + 2) = 9/11 \rightarrow 11x + 22 = 9y + 18 \rightarrow 11x - 9y = -4$

$(x + 3)/(y + 3) = 5/6 \rightarrow 6x + 18 = 5y + 15 \rightarrow 6x - 5y = -3$

$\times 5$ and $\times 9$: $55x - 45y = -20$; $54x - 45y = -27$

Subtract: $x = 7$

$6(7) - 5y = -3 \rightarrow 5y = 45 \rightarrow y = 9$

Fraction = 7/9

1 (vi) **Adding 1 to the numerator and subtracting 1 from the denominator gives 1. Adding 1 only to the denominator gives 1/2. Find the fraction.**

✔ ANSWER

$(x + 1)/(y - 1) = 1 \rightarrow x - y = -2$

$x/(y + 1) = 1/2 \rightarrow 2x - y = 1$

Subtract: $x = 3$

$y = x + 2 = 5$

Fraction = 3/5

1 (vii) **Five years ago, Nuri was thrice as old as Sonu. Ten years later, Nuri will be twice as**

old as Sonu. Find their ages.

✔ ANSWER

Nuri x years, Sonu y years

$x - 5 = 3(y - 5) \rightarrow x - 3y = -10$

$x + 10 = 2(y + 10) \rightarrow x - 2y = 10$

Subtract: $y = 20$

$x = 10 + 40 = 50$

Nuri 50 years, Sonu 20 years

1 (viii) **Sum of the digits of a two-digit number is 9. Nine times the number is twice the reversed number. Find the number.**

✔ ANSWER

Tens digit x, units digit y: $x + y = 9$

$9(10x + y) = 2(10y + x) \rightarrow 90x + 9y = 20y + 2x \rightarrow 88x = 11y \rightarrow y = 8x$

$x + 8x = 9 \rightarrow x = 1, y = 8$

Number = 18 ($9 \times 18 = 162 = 2 \times 81$)

1 (ix) **Meena withdrew ₹2000 in ₹50 and ₹100 notes, 25 notes in all. How many of each?**

✔ ANSWER

x notes of ₹50, y notes of ₹100

$x + y = 25$

$50x + 100y = 2000 \rightarrow x + 2y = 40$

Subtract: $y = 15$, $x = 10$

10 notes of ₹50 and 15 notes of ₹100

1 (x) **Library: fixed charge for the first three days plus a charge per extra day. 7 days cost ₹27, 5 days cost ₹21. Find both charges.**

✔ ANSWER

Fixed ₹x, ₹y per extra day

7 days = 3 + 4 extra: x + 4y = 27

5 days = 3 + 2 extra: x + 2y = 21

Subtract: 2y = 6 → y = 3

x = 21 − 6 = 15

Fixed charge ₹15, ₹3 for each extra day

2 **10 students took part in a quiz. The number of girls is 4 more than the number of boys. Find the numbers graphically.**

✔ ANSWER

Girls x, boys y

x + y = 10 → points (10, 0), (5, 5), (7, 3)

x − y = 4 → points (4, 0), (6, 2), (7, 3)

Draw both lines; they meet at (7, 3).

7 girls and 3 boys

3 (i) **Use the ratios: 5x − 4y + 8 = 0; 7x + 6y − 9 = 0**

✔ ANSWER

$a_1/a_2 = 5/7$, $b_1/b_2 = -4/6 = -2/3$

$5/7 \neq -2/3$

Lines intersect at a point

3 (ii) **$9x + 3y + 12 = 0$; $18x + 6y + 24 = 0$**

✔ **ANSWER**

$9/18 = 1/2$, $3/6 = 1/2$, $12/24 = 1/2$

All equal

Lines are coincident

3 (iii) **$6x - 3y + 10 = 0$; $2x - y + 9 = 0$**

✔ **ANSWER**

$6/2 = 3$, $-3/-1 = 3$, $10/9$

$a_1/a_2 = b_1/b_2 \neq c_1/c_2$

Lines are parallel

4 (i) **Do these have solutions? If yes, find them graphically: $x + y = 5$, $2x + 2y = 10$**

✔ **ANSWER**

$1/2 = 1/2 = 5/10 \rightarrow$ coincident lines

Infinitely many solutions: every point on $x + y = 5$, e.g. (0, 5), (2, 3), (5, 0)

4 (ii) **$x - y = 8$, $3x - 3y = 16$**

✔ **ANSWER**

$1/3 = 1/3$, but $8/16 = 1/2 \neq 1/3$

Parallel lines

No solution

4 (iii) **2x + y − 6 = 0, 4x − 2y − 4 = 0**

✔ ANSWER

2/4 = 1/2, 1/(−2) = −1/2 → not equal → unique solution

Line 1: (0, 6), (3, 0); Line 2: (0, −2), (1, 0)

The lines meet at (2, 2).

x = 2, y = 2 (check: 4 + 2 − 6 = 0; 8 − 4 − 4 = 0)

4 (iv) **2x − 2y − 2 = 0, 4x − 4y − 5 = 0**

✔ ANSWER

2/4 = 1/2, −2/−4 = 1/2, but −2/−5 = 2/5 ≠ 1/2

Parallel lines

No solution

5 **Half the perimeter of a rectangular garden is 36 m, and the length is 4 m more than the width. Find the dimensions.**

✔ ANSWER

l + w = 36, l − w = 4

Add: 2l = 40 → l = 20

w = 16

Length 20 m, width 16 m

6 **Given 2x + 3y − 8 = 0, write another linear equation so that the pair gives (i) intersecting (ii) parallel (iii) coincident lines.**

💡 SAMPLE ANSWER

(i) **3x − 2y − 7 = 0** (2/3 ≠ 3/−2)

(ii) **4x + 6y − 12 = 0** (2/4 = 3/6 = 1/2 but −8/−12 = 2/3)

(iii) **4x + 6y − 16 = 0** (2/4 = 3/6 = −8/−16 = 1/2)

(Many other answers are possible.)

7 **Mahāvīrāchārya: 9 citrons and 7 wood-apples cost 107; 7 citrons and 9 wood-apples cost 101. Find the price of each.**

✔ ANSWER

9x + 7y = 107; 7x + 9y = 101

Add: 16x + 16y = 208 → x + y = 13

Subtract: 2x − 2y = 6 → x − y = 3

Add these: 2x = 16 → x = 8; y = 5

Citron = 8, wood-apple = 5

Special: the coefficients are interchanged, so adding and subtracting give simple equations.

8 **5 pencils and 7 pens cost ₹50; 7 pencils and 5 pens cost ₹46. Find the cost of each.**

✔ ANSWER

5x + 7y = 50; 7x + 5y = 46

Add: 12x + 12y = 96 → x + y = 8

Subtract: $2x - 2y = -4 \rightarrow x - y = -2$

$x = 3, y = 5$

Pencil ₹3, pen ₹5

9 **Find the height of the stool. (Picture: cat sitting on the stool, total height 85 cm; cat on the floor beside it, stool is 25 cm taller than the cat.)**

✔ ANSWER

Stool s cm, cat c cm

$s + c = 85$

$s - c = 25$

Add: $2s = 110 \rightarrow s = 55$ (cat = 30 cm)

Height of the stool = 55 cm

📝 End-of-Chapter Exercises

1 (i) **Line y = 3x passes through (0, 0) and (2, 6). Identify the slope and explain using the two points.**

✔ ANSWER

Rise = 6 − 0 = 6, run = 2 − 0 = 2

Slope = 6/2 = **3** (same as m in y = 3x)

1 (ii) **What is the y-intercept? Explain its significance.**

✔ ANSWER

y-intercept = 0: the line crosses the y-axis at the origin.

In the tank story: at x = 0 (start), the water level is 0 cm, i.e. the tank starts empty.

1 (iii) **Water level y (cm) after x minutes follows y = 3x. (a) How high after 5 minutes? (b) When will it reach 21 cm?**

✔ ANSWER

(a) y = 3 × 5 = **15 cm**

(b) 21 = 3x → x = **7 minutes**

1 (iv) **Without drawing, does (4, 10) lie on this line?**

✔ ANSWER

At x = 4, y = 3 × 4 = 12 ≠ 10

No, (4, 10) is not on the line.

1 (v) **Plot the point where this line meets x = 3. Explain.**

✔ ANSWER

On x = 3 every point has x = 3.

Put x = 3 in y = 3x: y = 9

Point (3, 9)

2 (i) **F = (9/5)C + 32. Draw the graph with Celsius on the x-axis and Fahrenheit on the y-axis.**

✔ ANSWER

Table: C = 0 → F = 32; C = 100 → F = 212; C = −40 → F = −40; C = 30 → F = 86

Plot (0, 32), (100, 212), (−40, −40) and draw the straight line through them.

Slope 9/5, y-intercept 32.

2 (ii) **30 °C in Fahrenheit?**

✔ ANSWER

F = 9/5 × 30 + 32 = 54 + 32

= 86 °F

2 (iii) **95 °F in Celsius?**

✔ ANSWER

95 = 9C/5 + 32

9C/5 = 63 → C = 63 × 5/9

C = 35 °C

2 (iv) **0 °C in Fahrenheit, and 0 °F in Celsius?**

✔ ANSWER

C = 0: F = 0 + 32 = **32 °F**

F = 0: 9C/5 = −32 → C = −160/9

≈ −17.8 °C

2 (v) **Is there a temperature that is numerically the same in both scales?**

✔ ANSWER

Put F = C = t: t = 9t/5 + 32

t − 9t/5 = 32 → −4t/5 = 32 → t = −40

Yes: −40 °C = −40 °F

3 **Solve graphically: 2x + y = 6, 2x − y − 2 = 0.**

✔ ANSWER

2x + y = 6: (0, 6), (3, 0), (1, 4)

2x − y = 2: (0, −2), (1, 0), (3, 4)

The lines meet at (2, 2).

x = 2, y = 2

4 **Find the point of intersection of the lines shown on the cover page.**

💡 SAMPLE ANSWER

Read two points on each line from the cover, write each line's equation (y = mx + d), then solve the pair by substitution or elimination.

The intersection is the point that satisfies both equations. (check the picture in your book)

5 **Give a formula to find the x-intercept of the line y = mx + c.**

✔ ANSWER

At the x-axis, $y = 0$: $0 = mx + c$

$x = -c/m$ ($m \neq 0$)

If $m = 0$ (and $c \neq 0$), the line is horizontal and does not meet the x-axis.

6 **Plan A: ₹50 + ₹0.20 per minute. Plan B: ₹30 + ₹0.30 per minute. When is A cheaper, when is B cheaper, and when are they equal?**

✔ ANSWER

Cost for t minutes: A = 50 + 0.2t, B = 30 + 0.3t

Equal: 50 + 0.2t = 30 + 0.3t → 20 = 0.1t → t = 200

At 200 min both cost ₹90.

Same cost at 200 minutes; Plan A is cheaper for more than 200 minutes; Plan B is cheaper for less than 200 minutes.

7 (i) **How many lines have a given slope?**

✔ ANSWER

$y = mx + d$ with fixed m and any d

Infinitely many (all parallel)

7 (ii) **How many lines have a given slope and pass through a given point?**

✔ ANSWER

The point fixes d, so **exactly one line**.

8 **For what values of p does 4x + py + 8 = 0; 2x + 2y + 2 = 0 have a unique solution?**

✔ ANSWER

Unique solution when $a_1/a_2 \neq b_1/b_2$

$4/2 \neq p/2 \rightarrow 2 \neq p/2 \rightarrow p \neq 4$

All values of p except p = 4

9 **Find a and b so that (a + b)x − 2by = 5a + 2b + 1; 3x − y = 14 has infinitely many solutions.**

✔ ANSWER

Need $(a + b)/3 = -2b/(-1) = (5a + 2b + 1)/14$

$(a + b)/3 = 2b \rightarrow a + b = 6b \rightarrow a = 5b$

$2b = (5a + 2b + 1)/14 \rightarrow 28b = 5a + 2b + 1 \rightarrow 26b = 25b + 1 \rightarrow b = 1$

$a = 5$

a = 5, b = 1 (check: 6x − 2y = 28 is 2 × (3x − y = 14))

10 **Find k so that x + 2y = 3; (k − 1)x + (k + 1)y = k + 3 are coincident lines.**

✔ ANSWER

Need $1/(k - 1) = 2/(k + 1) = 3/(k + 3)$

$1/(k - 1) = 2/(k + 1) \rightarrow k + 1 = 2k - 2 \rightarrow k = 3$

Check: $2/(3 + 1) = 1/2$ and $3/(3 + 3) = 1/2$ ✓

k = 3

11 (i) **Robot 1 starts at the origin, repeatedly 3 right then 4 up. Robot 2 starts at (10, 0), repeatedly 1 right then 2 up. Do the paths intersect? Where?**

💡 **SAMPLE ANSWER**

Robot 1 follows $y = (4/3)x$ (corners (3, 4), (6, 8), ...)

Robot 2 follows $y = 2(x - 10)$ (corners (11, 2), (12, 4), ...)

$4x/3 = 2x - 20 \rightarrow 20 = 2x/3 \rightarrow x = 30, y = 40$

(30, 40) is reached by Robot 1 after 10 moves and by Robot 2 after 20 moves.

Yes, at (30, 40).

11 (ii) **Robot 1 starts at (3, 0), repeatedly 5 right then 5 up. Robot 2 starts at (7, 0), repeatedly 5 right then 3 down. Do the paths intersect?**

💡 **SAMPLE ANSWER**

Robot 1 follows $y = x - 3$ (going up); Robot 2 follows $y = -3(x - 7)/5$ (going down).

The lines meet where $x - 3 = (21 - 3x)/5 \rightarrow 8x = 36 \rightarrow x = 4.5, y = 1.5$.

But Robot 2 only moves to the right of $x = 7$ and below the x-axis, while Robot 1 goes above it, so (4.5, 1.5) is not on Robot 2's path.

No, the paths do not cross (they only share the x-axis between (7, 0) and (8, 0) at the very start, if the first step-by-step moves are followed).

12 **Jacob's watch reads 'a' minutes after 2 o'clock; 15 minutes later it reads 'b' minutes after 3 o'clock; a = 6b. What time was it the second time?**

✔ ANSWER

2:a + 15 min = 3:b → a + 15 = 60 + b → a − b = 45

a = 6b → 5b = 45 → b = 9, a = 54

First look 2:54, second look **3:09**

13 **Sum of the digits of a two-digit number is 15. The reversed number exceeds it by 9. Find the number.**

✔ ANSWER

Tens x, units y: x + y = 15

(10y + x) − (10x + y) = 9 → 9y − 9x = 9 → y − x = 1

2y = 16 → y = 8, x = 7

Number = 78 (87 − 78 = 9)

14 **Cyclic quadrilateral ABCD: ∠A = (x + 7)°, ∠B = (y + 8)°, ∠C = (3y + 23)°, ∠D = (4x + 12)°. Find all angles.**

✔ ANSWER

Opposite angles of a cyclic quadrilateral add to 180°.

A + C: x + 7 + 3y + 23 = 180 → x + 3y = 150

B + D: y + 8 + 4x + 12 = 180 → 4x + y = 160

y = 160 − 4x → x + 480 − 12x = 150 → 11x = 330 → x = 30, y = 40

∠A = 37°, ∠B = 48°, ∠C = 143°, ∠D = 132°

15 **A train takes 6 hours less if its speed is 6 km/h more, and 6 hours more if its speed is 4 km/h less. Find the distance and speed.**

✔ ANSWER

Speed v km/h, time t h, distance = vt

(v + 6)(t − 6) = vt → −6v + 6t − 36 = 0 → t − v = 6

(v − 4)(t + 6) = vt → 6v − 4t − 24 = 0 → 3v − 2t = 12

t = v + 6 → 3v − 2v − 12 = 12 → v = 24, t = 30

Distance = 24 × 30 = 720

Speed 24 km/h, distance 720 km

16 **A father's age equals the sum of his four children's ages. After 20 years, the children's**

ages will add up to twice the father's age. Find the father's age.

✔ ANSWER

Father x, children's total y: x = y

After 20 years each of 4 children adds 20: y + 80 = 2(x + 20)

x + 80 = 2x + 40 → x = 40

Father is 40 years old

🎉 **Chapter 13 done! Revise the answers once more tomorrow.**